AoI and Energy-Aware Data Collection for IRS-Assisted UAV-IoT Networks Under Jamming

Peng Wang, *Graduate Student Member, IEEE*, Kai Liu, *Member, IEEE*, Yaodong Ma, *Graduate Student Member, IEEE*, and Qiang Gao, *Senior Member, IEEE*

***Abstract*—With the rapid popularity of autonomous aerial vehicles (AAV)-Internet of Things (IoT) networks, timely and energy-efficient data collection is a critical issue. The strong Line-of-Sight (LoS) communication of UAVs makes them more susceptible to jamming attacks, resulting in increased communication latency and energy consumption (EC). In this article, the problem of jointly minimizing Age of Information (AoI) and EC in data collection under malicious jamming attacks is formulated by optimizing the diagonal phase shift matrix of the intelligent reflective surface (IRS), UAV trajectory, and the transmit powers of IoT devices. The formulated problem is solved by an alternating optimization (AO) scheme. Specifically, we first obtain the closed-form solution of the IRS diagonal phase shift matrix by using the quantitative passive beamforming method. Then, the UAV trajectory is optimized by the variable parameter particle swarm annealing (VPPSA) algorithm, and the suboptimal solution of the transmit powers of IoT devices is obtained by the successive convex approximation (SCA) algorithm. Extensive simulation results demonstrate that the proposed algorithm outperforms the benchmark schemes in terms of AoI and EC.**

***Index Terms*—Age of Information (AoI), anti-jamming, data collection, energy consumption (EC), intelligent reflecting surface (IRS), unmanned aerial vehicle (UAV).**

I. Introduction

With the advantages of Line-of-Sight (LoS) communication, low costs, and high-environmental adaptability, unmanned aerial vehicle (UAV)-assisted data collection obtains numerous applications in UAV-Internet of Things (IoT) networks, such as disaster relief [1], monitoring [2], and automatic driving [3]. In general, these applications require stringent information timeliness, which can be indicated by the Age of Information (AoI). In addition, due to the limited energy of UAVs and IoT devices, energy consumption (EC) needs to be considered. Meanwhile, due to the LoS communication characteristic, the UAV-IoT networks are more vulnerable to jammers. Therefore, jointly minimizing the AoI and EC in UAV-IoT networks under malicious jamming is a very important problem [4].

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The authors are with the School of Electronics and Information Engineering and the State Key Laboratory of CNS/ATM, Beihang University, Beijing 100191, China (e-mail: wangpeng572@buaa.edu.cn; liuk@buaa.edu.cn; yaodongma@buaa.edu.cn; gaoqiang@buaa.edu.cn).

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The outdated information not only causes resource wastage but also leads to wrong judgment, leading to accidents and major disasters [5]. Therefore, to maintain the freshness of data collection, a new concept of AoI was first proposed in [6], which is defined as the time elapsed since the data is generated to the data is successfully decoded by the receiver. Subsequently, the problem of minimizing AoI has been widely studied in IoT devices [7], UAV networks [8], [9], and UAV-IoT networks [10], [11]. Wu et al. [8] minimized the average AoI by optimizing the UAV trajectory between its sensing position and transmission position. Ferdowsi et al. [9] aimed to minimize the normalized weighted sum of AoI (NWAoI) in the uplink data collection link by optimizing the UAV's trajectory between fixed nodes. In [10], the UAV trajectory optimization process is modeled as a constrained Markov decision process (CMDP) to minimize AoI in data collection. Qin et al. [11] utilized the queuing theory to analyze the effect of the number of IoT devices on the peak AoI (PAoI) in data collection. However, the above works did not consider the effect of the transmit powers of IoT devices on AoI.

The works [12], [13], [14] minimized the system AoI by optimizing the UAV trajectory and transmit power of the UAV or IoT devices, ensuring the timeliness of the information transmission. Jiang et al. [12] used the block coordinate descent (BCD) method to alternately optimize the UAV trajectory, the transmission power of the device, and the data upload time to achieve an average AoI minimization. In [13], the average AoI is minimized by jointly optimizing the UAV trajectory, the power of the power equipment, and the data acquisition time in a multi-UAV-assisted wireless power transmission scenario. Liu et al. [14] minimized the average AoI of ground nodes by optimizing the UAV trajectory and the power of multiple ground nodes. However, they ignored that the UAV and IoT devices are both energy-limited, thus the EC of UAVs and IoT devices also need to be considered.

In [15], the average AoI and transmitting energy-weighted sums is jointly minimized by optimizing the UAV flight speed, hovering locations, and bandwidth allocation of data collection in cellular networks using deep neural network approaches. Diao et al. [16] proposed a location and status update strategy optimization problem for UAV-flight-hovering mode, and further balanced the average AoI and average power consumption of UAV and IoT devices by optimizing the UAV trajectory. In [17], the system EC is minimized under the heterogeneous average PAoI constraint by optimizing the offloading strategy of the device and transmit power as well as the UAV trajectory. Yang et al. [18] proposed the mobile sensing nodes deployment problem in the UAV-aided sensed data collection networks (SDCNs), and realized the minimization of AoI and EC. However, the above works [15], [16], [17], [18] solved the problem of AoI and EC in UAV-IoT networks, without considering the issue of securing transmission under the threat of jamming attacks.

To ensure secure transmission, anti-jamming schemes should also be considered in the data collection processes for UAV-IoT networks. Specifically, Yang et al. [19] presented a distributed fault-tolerant communication algorithm to minimize the expected average PAoI in digital twin edge networks (DITENs) under jamming. Ma et al. [20] minimized the AoI violation probability in the jamming and eavesdropping environment by optimizing the transmit powers of IoT devices. Still, works [19], [20] did not consider the optimization of EC under jamming attacks. Ma et al. [21] deduced the intensity and probability density function of the jamming signal based on the received jamming in the case of a certain transmitted signal, and proposed a deep deterministic policy gradient (DDPG)-based framework to acquire the power control strategy in real-time, to improve the energy efficiency of the network, but they did not consider the optimization of AoI under jamming attacks. In [22], successive convex approximation (SCA) and BCD methods are used to optimize the trajectory of UAV and transmit power of nodes under malicious jamming and achieve the goal of maximizing the transmission rate. Wang et al. [23] maximized the interactive reward region (IRR) by optimizing the trajectory and transmit power of the UAV. However, these works [19], [20], [21], [22], [23] did not consider the low-probability LoS link in complex scenarios, which makes the optimization of AoI and EC more challenging during data collection.

As a revolutionary new technology, intelligent reflecting surface (IRS) can greatly enhance the probability of LoS links by adjusting the phase shift of passive reflecting elements. Therefore, LoS communication between UAV and IoT devices can be better secured by using IRS-assisted communication. In [24], the AoI and transmission data rate are optimized in aerial IRS-assisted IoT networks through the joint transmission scheduling, UAV location, and IRS phase shift matrix design. In [25], the UAV-IRS is employed to minimize the average sum of AoI (ASoA) in the network. Asim et al. [26] used clustering, difference, and greedy algorithms to minimize the overall cost of the UAV, including the EC, task completion time, and maintenance cost, by optimizing the UAV trajectory and IRS phase shift. Cai et al. [27] utilized a deep neural network method to minimize the average total system EC by optimizing the 3-D trajectory of the UAV and the phase of the IRS. However, none of the above works [24], [25], [26], [27] considered the joint optimization of AoI and EC under malicious jamming.

In summary, recent works have not studied the problem of timeliness and limited energy for uplink data collection in the presence of malicious jamming. In this article, AoI is used to characterize the problem of timeliness and EC is used to characterize the problem of limited energy. Furthermore, the optimization problems of the diagonal phase shift matrix of the IRS, the UAV trajectory, and the transmit powers of the IoT devices are considered to minimize the AoI and EC. Since these problems are coupling nonconvex problems, traditional algorithms are difficult to solve, we combine the traditional algorithm with the heuristic algorithm [28] to solve this problem. The main contributions of this article are summarized as follows.

1) We employ IRS to facilitate data collection in the UAV-IoT system under malicious jamming. First, the IRS is adopted to provide better LoS communication conditions for the network. Second, jammers are introduced to affect the signal-to-interference-and-noise ratio (SINR) of the received signal of the UAV. Finally, we present a joint optimization scheme to minimize EC and AoI, which includes optimizing the diagonal phase shift matrix of the IRS, designing the UAV trajectory, and adjusting the transmit powers of IoT devices.
2) To solve the above nonconvex problem, we transform it into three subproblems for solving based on an alternating optimization (AO) scheme. First, the closed solution of the IRS diagonal phase shift matrix is obtained by the quantitative passive beamforming method. Then, a joint particle annealing and convex approximation (JPACA) algorithm is proposed to solve the coupling problem between the UAV trajectory and transmit powers of IoT devices, which includes the variable parameter particle swarm annealing (VPPSA) algorithm and the SCA algorithm.
3) In the JPACA algorithm, the VPPSA algorithm proposed in this article is used to solve the trajectory optimization problem of the UAV. Then, the transmit powers optimization problem of IoT devices is solved by the SCA algorithm. The VPPSA algorithm is based on the particle swarm optimization (PSO) algorithm to make each particle represent the position information of all stopping points. Second, the simulated annealing (SA) algorithm is introduced to solve the nonconvexity of the problem, and then the updating method of inertia weight and the annealing strategy is optimized to balance the proportion of the algorithm in the global and local search.
4) Extensive simulation experiments evaluate the superiority of the algorithm. Specifically, the simulation results show that the method exhibits better performance than the benchmark scheme in terms of AoI, EC, and anti-jamming. In addition, the algorithm also outperforms other algorithms when the number and distribution of IoT devices change.

The remainder of this article is organized as follows. In Section II, we present the system model and formulate the problem. Section III describes the proposed algorithm. Section IV gives the simulation results. Finally, Section V summarizes this article.

II. System Model and Problem Formulation

In the considered uplink data collection system, the UAV collects data from multiple ground IoT devices aided by a

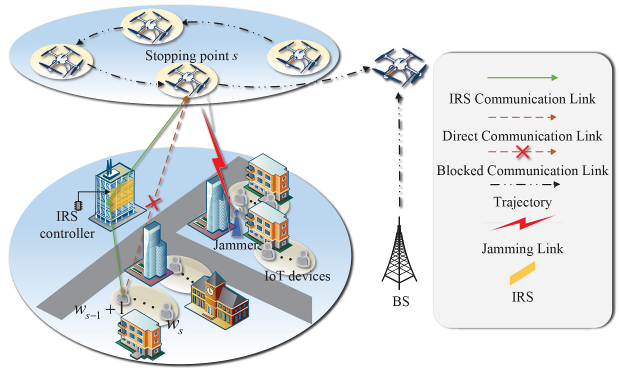

Fig. 1. Illustration of the uplink UAV-IoT communication network.

passive IRS in the threat of the malicious jammer, as shown in Fig. 1. IoT devices transmit data to the UAV via direct links and IRS-reflected links during data collection, with system optimization to minimize the AoI in the network, the EC of UAVs and IoT devices, and the effect of malicious jammer on the information. All communication nodes (i.e., the BS, jammer, IRS, and IoT devices) are located in 3-D Cartesian coordinates, in which the positions of the BS are denoted as $\mathbf{q}_B = [x_B, y_B, H_B]$. The set of randomly placed jammers can be denoted as $\mathcal{J} \triangleq \{1, 2, \ldots, J\}$, the coordinates of the jth jammer can be denoted as $\mathbf{q}_j = [x_j, y_j, H_j], j \in \mathcal{J}$. Similar to [29], we assume that the position of IoT devices follows a Poisson distribution and IoT devices are denoted by sets $\mathcal{K} \triangleq \{1, 2, \ldots, K\}$. Then, the position of kth IoT device is represented as $\mathbf{q}_k = [x_k, y_k, 0], k \in \mathcal{K}$. Moreover, similar to [26] and [27], the IRS is mounted on buildings parallel to the XoZ plane. It has a uniform linear array (ULA) with $M = M_x M_y$ reflecting elements and its coordinate is denoted by $\mathbf{q}_I = [x_I, y_I, H_I]$. The UAV is flying at a fixed altitude H_0, and one flight period is T consisting of S stopping points. The stopping points set is denoted by $\mathcal{S} \triangleq \{1, 2, \ldots, S\}$. Therefore, the coordinate of the sth stopping point is $\mathbf{q}_s = [x_s, y_s, H_0]$, $s \in \mathcal{S}$.

Similar to [29], considering the influence of bandwidth and the amount of data that IoT devices need to transmit, it is limited that each stopping point can communicate with up to W IoT devices. According to [30] and [31], once the UAV reaches a stopping point, it transmits beacon message to W IoT devices which require data transmission. The beacon message includes the type of sensor nodes to be activated in response to the beacon, the data collection height of the UAV, and a threshold to limit the number of sensor nodes in the IoT devices. Therefore, IoT devices will transmit data to the UAV in a certain order according to the beacon message during the process of data collection. To avoid repeated collection of data from IoT devices in one cycle, according to the above communication model, we divide all IoT devices into S parts according to the number of stopping points that can be made and the distance of devices from BS, as shown in Fig. 1. Therefore, the set of IoT devices corresponding to the stopping point s can be represented as $\mathcal{K} \triangleq \{w_{s-1}+1, \ldots, w_s\}\ \ \forall s \in \mathcal{S}$. And we take it that $w_0 = 0$. In this way, after the device $w_{s-1}+1$ belonging to the stopping point s has sent the data, the UAV will notify the next device $w_{s-1}+2$ to send the data. Meanwhile, the UAV will also send the corresponding phase control information to the IRS. It can not only ensure the data collection of all IoT devices but also ensure that the data collection of each IoT device is only collected once.

A. Channel Model

In the considered system, we assume that the UAV, jammer, BS, and multiple IoT devices are equipped with an omnidirectional antenna. The channels between IRS and UAV, and that between IoT devices and IRS, follow a path loss and small-scale fading channel model.

The data of IoT devices is collected by direct reception and IRS reflection after the UAV has reached the stopping point. First, we know that the distance between the sth stopping point and IRS is

$$d_s^{I,U} = \sqrt{(x_s - x_I)^2 + (y_s - y_I)^2 + (H_0 - H_I)^2}. \tag{1}$$

Similarly, we can also get the distance between the kth IoT device and the sth stopping point of the UAV is $d_{s,k}^{U,E}$, the distance between the kth IoT device and the IRS is $d_k^{I,E}$. It can also be obtained that the distance between the jth jammer and the sth stopping point of the UAV is $d_{j,s}^{J,U}$.

LoS link and non-LoS (NLoS) link are considered in the design of ground-to-air communication for the UAV hovering at the stopping point. The LoS link probability is related to the environment, elevation, and transmission distance [32], thus the LoS communication probability at the stopping point s can be expressed as

$$P_{s,k}^{\text{LoS}} = \frac{1}{1 + \alpha \exp\bigl(-\beta(\theta_{s,k} - \alpha)\bigr)} \tag{2}$$

where α and β are constants determined by the environment, $\theta_{s,k} = \arctan(H_0/r_{s,k})$ is the elevation Angle between the kth IoT device and the UAV at the sth stopping point, and $r_{s,k} = \sqrt{(x_k - x_s)^2 + (y_k - y_s)^2}$ is the horizontal distance between the kth IoT device and the sth stopping point. The corresponding probability of NLoS communication is $P_{s,k}^{\text{NLoS}} = 1 - P_{s,k}^{\text{LoS}}$.

In addition, we assume that $\theta_{s,k}[m] = [0, 2\pi), m \in \mathcal{M} \triangleq \{1, \ldots, M\}$ is the phase shift of a single reflection unit, where m represents the mth reflection element. Therefore, the diagonal phase shift matrix of IRS can be expressed as $\Theta_{s,k} \triangleq \text{diag}\{e^{j\theta_{s,k}[1]}, \ldots, e^{j\theta_{s,k}[m]}, \ldots, e^{j\theta_{s,k}[M]}\}$. Due to the rare blockages in the air and the flexible deployment of the IRS, we assume that all channels are LoS channels in the considered system. On this basis, the channel gain between the IRS to the sth stopping point and the IRS to the kth IoT device can be expressed as

$$h_s^{I,U} = \sqrt{\frac{\gamma_1}{\left(d_s^{I,U}\right)^2}}\Bigl[1, \ldots, e^{-j\frac{2\pi}{\lambda}d\left((m_x-1)\phi_{s,x}^{I,U}+(m_y-1)\phi_{s,y}^{I,U}\right)}, \ldots, e^{-j\frac{2\pi}{\lambda}d\left((M_x-1)\phi_{s,x}^{I,U}+(M_y-1)\phi_{s,y}^{I,U}\right)}\Bigr]^T \tag{3}$$

$$h_k^{I,E} = \sqrt{\frac{\gamma_1}{\left(d_k^{I,E}\right)^2}}\Big[1, \ldots, e^{-j\frac{2\pi}{\lambda}d\left((m_x-1)\phi_{k,x}^{I,E}+(m_y-1)\phi_{k,y}^{I,E}\right)} \ldots, e^{-j\frac{2\pi}{\lambda}d\left((M_x-1)\phi_{k,x}^{I,E}+(M_y-1)\phi_{k,y}^{I,E}\right)}\Big]^T \tag{4}$$

where γ_1 represents the path loss out of 1 m, the right side of the formula represents the response of the array to IRS. $\phi_{s,x}^{I,U} = [(x_I - x_s)/d_s^{I,U}]$ and $\phi_{s,y}^{I,U} = [(y_I - y_s)/d_s^{I,U}]$ represent the cosine of the Angle of Departure (AoD) from IRS to the sth stopping point. λ is the carrier wavelength and d is the IRS element separation distance. $\phi_{k,x}^{I,E} = [(x_k - x_I)/d_k^{I,S}]$ and $\phi_{k,y}^{I,E} = [(y_k - y_I)/d_k^{I,S}]$ represents the cosine of the Angle of Arrival (AoA) from the kth IoT device to IRS.

Thus, the cascaded IoT device-IRS-UAV channel can be represented as

$$h_{s,k}^{\text{EIU}} = \left(h_s^{I,U}\right)^H \Theta_{s,k} h_k^{I,E} \tag{5}$$

where H stands for conjugate transpose. Further, the direct link between IoT devices and the UAV and the link between the UAV and the jammer also undergo path loss and small-scale fading. The direct channel gain between the kth IoT devices and the UAV at the sth stopping point can be expressed as

$$h_{s,k}^{U,E} = \sqrt{\frac{\gamma_2}{\left(d_{s,k}^{U,E}\right)^2}} e^{-j\frac{2\pi d_{s,k}^{U,E}}{\lambda}} \tag{6}$$

where γ_2 represents the path loss out of 1 m.

Similar to [29] and [33], considering that NLoS communication will greatly reduce the communication quality, we assume that when the LoS link between the IoT device and the UAV is not available, the signal transmitted by the IoT device can only be reflected and enhanced to the UAV by the IRS. Therefore, this article considers that the channel gain under the LoS communication condition is $h_{s,k}^{U_{\text{LoS}}} = h_{s,k}^{\text{EIU}} + h_{s,k}^{U,E}$. In contrast, the channel gain under the NLoS communication condition is $h_{s,k}^{U_{\text{NLoS}}} = h_{s,k}^{\text{EIU}}$. Therefore, the final channel gain can be expressed as

$$h_{s,k}^{U} = P_{s,k}^{\text{LoS}} h_{s,k}^{U_{\text{LoS}}} + P_{s,k}^{\text{NLoS}} h_{s,k}^{U_{\text{NLoS}}} = h_{s,k}^{\text{EIU}} + P_{s,k}^{\text{LoS}} h_{s,k}^{U,E}. \tag{7}$$

The jammer sends jamming signals to reduce the SINR received by the UAV. Similar to [33], since the jammer is placed in building shadows to remain inconspicuous, we assume that the high-rise building blocks the communication link between the jammer and the IRS, thus the jammer affects the communication effect by directly jamming with the UAV. Therefore, we can get that the direct channel gain between the UAV at the sth stopping point and the jth jammer is

$$h_{j,s}^{J,U} = \sqrt{\frac{\gamma_2}{\left(d_{j,s}^{J,U}\right)^2}} e^{-j\frac{2\pi d_{j,s}^{J,U}}{\lambda}}. \tag{8}$$

Above all, the transmission rate between the k-IoT device and the s-stopping point can be obtained as

$$R_{sk} = \hat{B}\log_2\left(1 + \frac{p_k |h_{s,k}^U|^2}{\sum_{j=1}^J p_j |h_{j,s}^{J,U}|^2 + \sigma_s^2}\right) \tag{9}$$

where $\hat{B}$ is the system bandwidth, p_k is the transmit power of the kth IoT device, p_j is the fixed transmit power of the jth jammer, and σ_s^2 is the additive white Gaussian noise (AWGN) at the stopping point s. To simplify the model, we assume that the jammers have the same transmit power, i.e., $p_1 = \cdots = p_j = \cdots = p_J = P_J$ and the same AWGN at each stopping point, i.e., $\sigma_1^2 = \cdots = \sigma_s^2 = \cdots = \sigma_S^2 = \sigma^2$.

B. UAV Trajectory Model

It is assumed that the UAV begins to collect data from the BS at a constant speed of V. It hovers to collect data after reaching the stopping point and then proceeds to the next stopping point to collect more data. After collecting all the necessary data, the UAV returns to the BS, thus we set the initial point of the UAV as $\mathbf{q}_0$ and the endpoint of the UAV as $\mathbf{q}_{S+1}$, there is

$$\mathbf{q}_0 = \left[x_B, y_B, H_0\right] \tag{10a}$$

$$\mathbf{q}_{S+1} = \mathbf{q}_0. \tag{10b}$$

Then, the total flight time of the UAV is

$$T_{\text{fly}} = \frac{\sum_{s=1}^{S+1} \|\mathbf{q}_s - \mathbf{q}_{s-1}\|_2}{V} \tag{11}$$

where $\|\cdot\|_2$ denotes the Euclidean norm. Since each stopping point is connected to no more than W IoT devices, the limit of bandwidth can be expressed as

$$1 \le w_s - w_{s-1} \le W \quad \forall s \in \mathcal{S} \tag{12}$$

Then, to ensure smooth UAV movement, the number of stopping points is limited as follows:

$$S = \left\lceil \frac{K}{W} \right\rceil \tag{13}$$

where $\lceil\cdot\rceil$ denotes the ceiling function. To ensure that the data of each IoT device can be successfully uploaded, it also needs to be met

$$w_S = K. \tag{14}$$

Finally, the hovering time of the UAV should be

$$T_{\text{hov}} = \sum_{s=1}^{S} \sum_{k=w_{s-1}+1}^{w_s} \frac{D_k}{R_{sk}} \tag{15}$$

where D_k represents the size of data that the kth IoT device needs to send. For ease of analysis, we consider that the data sent by each IoT device is of the same size.

C. Energy Consumption Model

The EC of UAV is mainly divided into two parts, one is propulsion EC $E_p = p_f T_{\text{fly}}$ and the other is hovering EC $E_{\text{hov}} = p_h T_{\text{hov}}$. Following [34], the propulsion power consumption of the UAV for horizontal movement is the function of speed V, thus when the UAV's flight speed is considered to be constant, its propulsion power is also a definite value. Compared with these two items, the signal transmission EC of UAV is relatively small, thus it is ignored. In practice, the IRS is usually mounted on the building exterior which is accessible

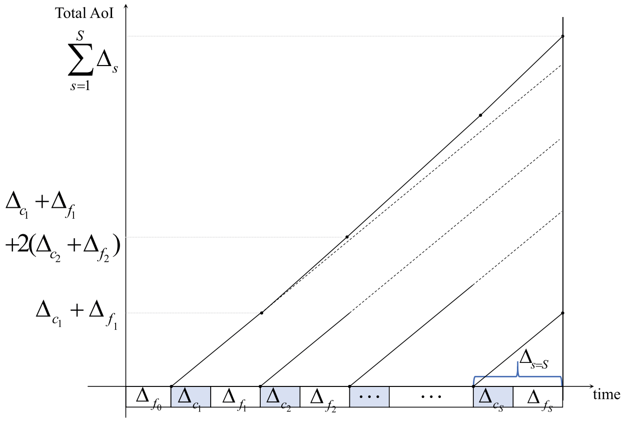

Fig. 2. Examples of the evolution of AoI.

to the energy source. Besides, the IRS is nearly passive and its operation power is a constant which is much lower than that of the EC of the UAV. Therefore, we ignore the IRS power consumption in the considered system. The EC of IoT devices are $E_K = \sum_{s=1}^{S}\sum_{k=w_{s-1}+1}^{w_s}(D_k/R_{sk})p_k$. So the total system EC is

$$E = E_p + E_{\text{hov}} + \omega_1 E_K \tag{16}$$

where ω_1 is the compensation factor because the smaller IoT device power causes it not to be at the same level as the UAV's propulsion and hovering EC. And p_f and p_h are the propulsion power and hovering power of the UAV, respectively.

D. AoI Model

As shown in Fig. 2, $\triangle_{f_0}$ is the time that the UAV flies from BS to the first stopping point. Since the UAV sends a beacon message to IoT devices in a short time compared to the upload time, the time for sending a beacon message can be ignored. Then, the generation time of the data of the IoT devices at the stopping point s can be regarded as the time when the UAV reaches the stopping point s. Specifically, the $\triangle_s$ is the AoI of data at the stopping point s and $\triangle_{c_s}= T_{\text{hov},s} = \sum_{k=w_{s-1}+1}^{w_s}(D_k/R_{sk})$ is the time that the UAV collects data at stopping point s. It is worth noting that, this article assumes that the UAV transmits the data to the BS after collecting the data from all IoT devices to reduce the EC of the UAV [35], [36], [37], [38], thus the $\triangle_{f_s}= T_{\text{fly},s} = ([\|\mathbf{q}_{s+1}-\mathbf{q}_s\|_2]/V)$ is the time that the UAV flies from stopping point s to stopping point $s+1$ and $\triangle_{f_S}$ is the time that the UAV flies from stopping point S to BS. For example, when the UAV collects the data of the second stopping point, the AoI of the first stopping point data is still increasing, i.e., $\triangle_{s=1}+\triangle_{s=2}= \Delta_{c_1}+\Delta_{f_1}+2(\Delta_{c_2}+\Delta_{f_2})$. Thus, the AoI of data collection at the stopping point s at time t in the general case can be defined as

$$\triangle_s = \begin{cases} 0, & t < t_s \\ t - t_s, & t \geq t_s \end{cases} \tag{17}$$

where t_s represents the data generation time of the IoT devices of the stopping point s. It is worth noting that, different from the traditional definition of delay [39], [40], AoI represents the time from the data collected by the UAV one-by-one to that it is finally decoded at the BS. Therefore, when the UAV transmits all data to the BS, i.e., $t = t_B$, the AoI of the data at the stopping point s is

$$\begin{aligned}\triangle_s = t_B - t_s &= \triangle_{c_s} + \triangle_{f_s} + \triangle_{c_{s+1}} + \triangle_{f_{s+1}} + \cdots + \triangle_{c_S} + \triangle_{f_S} \\ &= \frac{\sum_{i=s}^{S}\|\mathbf{q}_{i+1}-\mathbf{q}_i\|_2}{V} + \sum_{j=s}^{S}\sum_{k=w_{j-1}+1}^{w_j}\frac{D_k}{R_{jk}}\end{aligned} \tag{18}$$

where t_B represents the time when the UAV transmits the data to the BS. Subsequently, BS processes the data to determine whether the data is outdated, valid, etc. To simplify the model, we omit the problem of retransmission. Due to the transmission of beacon message, the issues of the data collection frequency can be ignored. Therefore, the total AoI of the system is

$$\text{AoI} = \sum_{s=1}^{S}\triangle_s\ . \tag{19}$$

E. Problem Formulation

The communication system includes IRS, UAV, jammer, IoT devices, and others. In this case, we hope to balance the relationship between EC and AoI by optimizing the diagonal phase shift matrix $\Theta_{s,k}$ of IRS, the stopping point $\mathbf{q}_s$ of the UAV, and the transmit powers p_k of IoT devices, while achieving anti-jamming and minimizing EC and AoI. Therefore, we weighted the two to obtain a new metric $\text{EAoI} = \omega_2\zeta\,\text{AoI} + (1-\omega_2)\text{E}$, in which $0 \leq \omega_2 \leq 1$ is the weight coefficient, ζ is the compensation factor of AoI, because AoI and EC are at different levels. The relationship between them can be better balanced after adding the compensation factor. So the optimization problem can finally be expressed as

$$\begin{aligned}(\mathcal{P}_0):\ \min_{\Theta_{s,k},\{p_k\},\{\mathbf{q}_s\}}\ \text{EAoI} = &\frac{\omega_2\zeta}{V}\sum_{s=1}^{S}\sum_{i=s}^{S}\|\mathbf{q}_{i+1}-\mathbf{q}_i\|_2 \\ &+ \omega_2\zeta D\sum_{s=1}^{S}\sum_{j=s}^{S}\sum_{k=w_{j-1}+1}^{w_j}\frac{1}{R_{jk}} \\ &+ (1-\omega_2)p_f T_{\text{fly}} + (1-\omega_2)p_h T_{\text{hov}} \\ &+ \omega_1(1-\omega_2)D\sum_{s=1}^{S}\sum_{k=w_{s-1}+1}^{w_s}\frac{p_k}{R_{sk}}\end{aligned} \tag{20a}$$

$$\text{s.t.}\quad \theta_{s,k}[m] \in [0, 2\pi),\ m \in \mathcal{M} \tag{20b}$$

$$0 < p_k \leq p_k^{\max}\quad \forall k \in \mathcal{K} \tag{20c}$$

$$0 \leq x_s \leq x_s^{\max}\quad \forall s \in \mathcal{S} \tag{20d}$$

$$0 \leq y_s \leq y_s^{\max}\quad \forall s \in \mathcal{S} \tag{20e}$$

$$(10), (12), (13), (14)$$

where (10) specifies the starting point and ending point of the UAV. Equation (12) gives the restrictions on IoT devices' communication. Equations (13) and (14) ensure the number of UAV stopping points, so that the UAV can still collect all IoT devices' data even when the number of communications is limited. Equation (20c) imposes limits on the transmit power of IoT devices. Equations (20d) and (20e) denote the range of values of x_s and y_s, respectively, where $x_{\max}$ and $y_{\max}$ are the maximum values of x_s and y_s, respectively.

The formulated problem $\mathcal{P}_0$ is a multiobjective traveling salesman problem (MOTSP), which aims to anti-jamming and minimize EC and AoI while combining the determination of stopping points with the trajectory planning problem of the UAV. The traditional TSP [41] involves finding a minimum-cost tour (i.e., the total length of the tour is minimum) that travels each region exactly once for a collection of compact areas before returning to the initial departure point. However, our formulated problem not only considers multiple optimization objectives but also the presence of the jammer and the cost spent at each stopping point and IoT device. At the same time, $\mathcal{P}_0$ is also an NP-hard problem, thus we will divide it into three subproblems to solve in the next section.

III. Proposed Solution

In general, there is no effective solution for the above coupled nonconvex optimization problem, therefore we propose to decompose it into tractable subproblems to solve it. Specifically, first, the IRS phase shift matrix $\Theta_{s,k}$ is optimized by the quantitative passive beamforming method, and the closed-form solution of the phase shift matrix is obtained. Then, the proposed JPACA algorithm is used to solve the suboptimal solution for the stopping point of the UAV and the transmit powers of IoT devices, where the location information of the optimized stopping point is obtained by the VPPSA method when the closed expression of the phase shift matrix and the transmit powers of IoT devices are given, the suboptimal solution for the transmit powers of IoT devices is obtained by the SCA method when the closed expression of the phase shift matrix and the location information of the stopping point are given. Finally, the two problems are solved iteratively [42]. Note that each variable block is updated sequentially during iteration, while others retain their last updated values. Denote T ($T = 1, 2, \ldots, T_{\max}$) as the index of, the outer iteration, n ($n = 1, 2, \ldots, N$) and c ($c = 1, 2, \ldots, \hat{c}$) are the indexes of the iterations of the VPPSA algorithm and the SCA algorithm.

A. Optimizing $\Theta_{s,k}$ for Given $\{\mathbf{q}_s\}$ and $\{p_k\}$

Before optimizing $\mathbf{q}_s$ and p_k, the optimization problem of the diagonal phase-shift matrix is independent of the first and third terms in (20a) and problem ($\mathcal{P}_0$) can be expressed by $\mathrm{EAoI}_{\Theta_{s,k}}$, which can be expressed as

$$(\mathcal{P}_{1.1}):\ \min_{\Theta_{s,k}}\ \mathrm{EAoI}_{\Theta_{s,k}} = \omega_2\zeta D\sum_{s=1}^{S}\sum_{j=s}^{S}\sum_{k=w_{j-1}+1}^{w_j}\frac{1}{R_{jk}} + (1-\omega_2)p_h D\sum_{s=1}^{S}\sum_{k=w_{s-1}+1}^{w_s}\frac{1}{R_{sk}} + \omega_1(1-\omega_2)D\sum_{s=1}^{S}\sum_{k=w_{s-1}+1}^{w_s}\frac{p_k}{R_{sk}}$$
$$\text{s.t. (20b).} \tag{21}$$

For that, we have

$$\frac{dR_{sk}}{dh_{s,k}^{\mathrm{EIU}}} = \varpi\frac{\frac{2p_k|h_{s,k}^{U}|}{\sum_{j=1}^{J}p_j|h_{j,s}^{J,U}|^2+\sigma_s^2}}{1+\frac{p_k|h_{s,k}^{U}|^2}{\sum_{j=1}^{J}p_j|h_{j,s}^{J,U}|^2+\sigma_s^2}} \geq 0 \tag{22}$$

where $\varpi = (\hat{B}/\ln 2)$. So it follows that R_{sk} is monotonic concerning $h_{s,k}^{\mathrm{EIU}}$.

Similar to [43], from the above formula, it can be seen that solving the minimization problem of ($\mathcal{P}_{1.1}$) can be converted to solving the problem of maximizing transmission rate, and then it can be concluded that the phase shift design problem is the problem of maximizing channel power gain. Therefore, the phase shift corresponding to an IoT device can be solved using the separation optimization method and the suboptimal solution of the problem can be obtained. The ($\mathcal{P}_{1.1}$) can be written as follows:

$$(\mathcal{P}_{1.2}):\ \max_{\Theta_{s,k}}\ \left|h_{s,k}^{\mathrm{EIU}}\right| \quad \text{s.t. (20b)} \tag{23}$$

where $h_{s,k}^{\mathrm{EIU}}$ can again be written as

$$h_{s,k}^{\mathrm{EIU}} = \frac{\gamma_1}{\left|d_s^{I,U}\right|\left|d_k^{I,E}\right|}\sum_{m=1}^{M}e^{j(\theta_{s,k}[m]+\varphi_m-\psi_m)} \tag{24}$$

where $\varphi_m = ([-2\pi d]/\lambda)((m_x-1)\phi_{k,x}^{I,E}+(m_y-1)\phi_{k,y}^{I,E})$, $\psi_m = ([-2\pi d]/\lambda)((m_x-1)\phi_{s,x}^{I,U}+(m_y-1)\phi_{s,y}^{I,U})$. So our goal is to find $\theta_{s,k}[m]$ that satisfies (20b) and maximizes $h_{s,k}^{\mathrm{EIU}}$.

Then, using the triangle inequality we can convert (24) into the following inequality:

$$\left|\sum_{m=1}^{M}e^{j(\theta_{s,k}[m]+\varphi_m-\psi_m)}\right| \leq \left|e^{j(\theta_{s,k}[1]+\varphi_1-\psi_1)}\right| + \cdots + \left|e^{j(\theta_{s,k}[m]+\varphi_m-\psi_m)}\right| + \cdots + \left|e^{j(\theta_{s,k}[M]+\varphi_M-\psi_M)}\right| = M. \tag{25}$$

Numerical analysis shows that the above equation holds if and only if $|\theta_{s,k}[m]+\varphi_m-\psi_m| = 0\ \forall m \in \mathcal{M}$. So we can obtain the optimization of problem ($\mathcal{P}_{1.2}$), the $|h_{s,k}^{\mathrm{EIU}}|$can take to the maximum when the phase shift of each element satisfies $\theta_{s,k}[m] = \psi_m - \varphi_m\ \forall m \in \mathcal{M}$. From this, the IRS can adjust the phase according to the stopping point and the IoT devices. On this basis, the UAV's trajectory and the IoT devices' transmit powers can be alternately optimized to further achieve anti-jamming and minimize EAoI.

B. Optimizing $\{\mathbf{q}_s\}$ for Given $\{p_k\}$ and $\Theta_{s,k}$

For given p_k and the closed expression of $\Theta_{s,k}$, the problem ($\mathcal{P}_0$) can be expressed by EAoI_q, which can be expressed as

$$(\mathcal{P}_2):\ \min_{\{\mathbf{q}_s\}}\ \mathrm{EAoI}_q = \frac{\omega_2\zeta}{V}\sum_{s=1}^{S}\sum_{i=s}^{S}\|\mathbf{q}_{i+1}-\mathbf{q}_i\|_2 + \omega_2\zeta D\sum_{s=1}^{S}\sum_{j=s}^{S}\sum_{k=w_{j-1}+1}^{w_j}\frac{1}{R_{jk}}$$

$$+(1-\omega_2)p_f T_{\text{fly}}+(1-\omega_2)p_h T_{\text{hov}}$$
$$+\omega_1(1-\omega_2)D\sum_{s=1}^{S}\sum_{k=w_{s-1}+1}^{w_s}\frac{p_k}{R_{sk}}$$
$$\text{s.t. (10), (12), (13), (14), (20d), (20e)} \quad (26)$$

where the result of EAoI_q represents the effect of the optimization of UAV trajectory on EAoI. The problem (26) is nonconvex and it is generally difficult to find an optimal solution. Then, a PSO-based algorithm can be designed to solve this problem. However, the standard PSO algorithm is proven to fall into the local optimal solution easily [28], [44]. Therefore, we propose a novel PSO-based algorithm to solve $\mathcal{P}_2$.

To make the problem easier to deal with, we first convert (12)–(14) to

$$\omega_s=\begin{cases} s\lceil\frac{K}{S}\rceil, & \text{if } s<S, s\in\mathcal{S}\\ K, & \text{others.}\end{cases} \quad (27)$$

Second, in the PSO algorithm, inertia weight w reflects the ability of particles to inherit the previous velocity. The larger the inertia weight, the more inclined to global search. The smaller the inertia weight, the more inclined to local search. However, in the general PSO algorithm, w usually takes a fixed constant value, which cannot properly balance the global search and local search capabilities of the algorithm, and the linearly decreasing inertia weight cannot properly handle the relationship between global search and local search when the population size is large [29], thus we adopt a nonlinear decreasing way to update the inertia weight w, as shown in the following formula:

$$w^n=\left(w_{\max}-(w_{\max}-w_{\min})\times\frac{n^3}{N^3}\right)\times\cos\left(\frac{n\pi}{2N}\right) \quad (28)$$

where $w_{\max}$ and w_{mim} are the maximum and minimum values of w, respectively, n is the current number of iterations, and N is the maximum number of iterations. And unlike the standard PSO algorithm, each particle p_i represents the horizontal coordinates of all S stopping points, and the maximum particle number P represents the population size, i.e., $p_i=\{(x_1^i,y_1^i),\dots(x_s^i,y_s^i),\dots,(x_S^i,y_S^i)\}$, as shown in Fig. 3. Thus, there are P particles in the algorithm, not $P\times S$ particles. This can not only consider the effect of UAV trajectory on EAoI from the perspective of all stopping points but also reduce the number of determinations of the SA algorithm, thereby reducing the time complexity. Based on the designed inertia weight update strategy, the update mode of position and velocity of each particle in the PSO algorithm can be expressed as

$$v_i^n=w^n\times v_i^{n-1}+c_1r_1\left(p_i^n-x_i^n\right)+c_2r_2\left(g^n-x_i^n\right) \quad (29)$$
$$x_i^{n+1}=x_i^n+v_i^n,\quad i=\{1,2,\dots,P\}\quad n=\{1,2,\dots,N\} \quad (30)$$

where the optimized weight coefficients can dynamically adjust the search range with the number of iterations, and the value of each particle fitness function is the value in the case of the stopping point distribution, i.e., $fit_i=\text{EAoI}_q$. Therefore, the above formula ensures that the PSO algorithm focuses on global search in the first and middle phases, and focuses on local search in the later phases, which improves the convergence speed of the algorithm.

In addition, to solve the problem that the conventional standard PSO algorithm easily falls into the local optimal solution, we introduce the SA algorithm [45] in the individual optimal solution update stage of the PSO algorithm, which ensures the convergence speed and increases the diversity of the population. As a greedy algorithm, the core idea of the SA algorithm is to find the global optimal solution by accepting a solution worse than the current one with probability p_{fit} to avoid falling into a local optimal solution, then p_{fit} can be denoted as

$$p_{\text{fit}}=\exp\left(\frac{\text{fit}_{\text{Pbest}}-fit_{\text{new}}}{\text{temp}^n}\right) \quad (31)$$

where fit_{new} and $\text{fit}_{\text{Pbest}}$ denote the value of EAoI_q corresponding to the current particle and the value of the optimal EAoI_q in the history of this particle, respectively, and temp^n is the current annealing temperature.

In this algorithm, we found that the traditional nonlinear cooling method, that is, the method of multiplying the initial temperature by the temperature attenuation coefficient, would cause the temperature to drop faster in the early stage. Therefore, this part also optimized the temperature updating strategy. We use a nonlinear method related to the number of iterations to update the annealing temperature and set a minimum annealing temperature. Thus, the temperature update strategy can be expressed as follows:

$$\text{temp}^n=T_{\text{ini}}-\left(\text{TE}_{\text{ini}}\times\frac{n^2}{N^2}\right)\times t\times\sin\left(\frac{n\pi}{2N}\right) \quad (32)$$

where TE_{ini} is the initial temperature and t is the value of adjusting the curve change rate.

To avoid the global optimal solution being affected by worse results, resulting in worse final output results of the algorithm, we only add the SA algorithm in the stage of the individual optimal solution, i.e., in the process of updating the individual optimal solution, the SA algorithm will determine whether to accept the stopping points with a worse EAoI as the individual optimal solution through (31) and (32). If a worse solution is received, the VPPSA algorithm continues to explore other possible stopping points, avoiding the algorithm falling into the local optimal solution. This not only overcomes the problem that the population diversity deteriorates in the iterative process of the PSO algorithm, but also avoids the situation of oscillation of the results.

Above all, the algorithm balances the relationship between flight time and hovering time by controlling the distance between UAV and IoT devices and between UAV and jammer, to reduce the effect of the jammer as much as possible and thus reduce EAoI. Each particle represents the information of all stopping points, and by optimizing the inertia weight parameters, combining the update of the individual optimal solution with the SA algorithm, which enhances the multiformity of solution space to make it skip out of the local optimal solution and improves the ability to search a global optimal solution, and overcomes the oscillation of the particle in the evolution process. In Algorithm 1, we summarize the

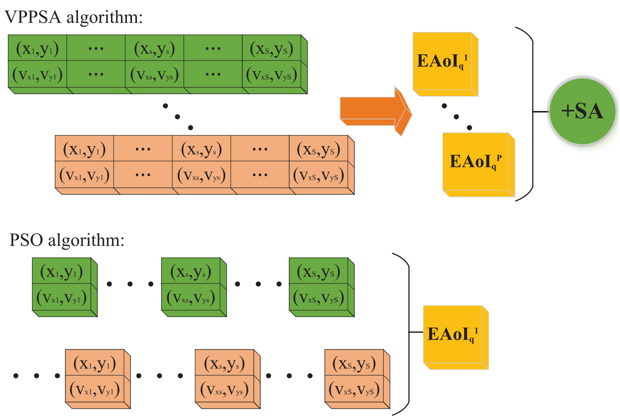

Fig. 3. VPPSA algorithm structure.

Algorithm 1: VPPSA Algorithm for Solving Problem $\mathcal{P}_2$

Initialization: iterations N, position x_i^n and velocity v_i^n of P particles, p_i^n and g^n, for $i = 1, 2, \ldots, P$, $n = 1, 2, \ldots, N$, $fit_i^n = \mathrm{EAoI}_q^i$;
for $n = 1$ to N **do**
  Define the update strategy of the inertia weight coefficient w
  for $i = 1$ to P **do**
    Update the velocity v_i^n and positions x_i^n of particles using (30) and (30);
    Calculate the fitness value by the $\mathcal{P}_2$;
    Check the position $x_i^n$and the speed v_i^n to correct the part that exceeds the threshold;
    if ($fit_{new} < fit_{Pbest}$) $\|$ $p_{fit} > th_{temp}$ **then**
      $p_i^n = x_i^n$ and $fit_{Pbest} = fit_{new}$
      if ($fit_{new} < fit_{Gbest}$) **then**
        $g^n = x_i^n$ and $fit_{Gbest} = fit_{new}$
      end if
    end if
  end for
  Renewal annealing temperature $temp^n$
end for

implementation details of the VPPSA algorithm, where $\mathrm{th}_{\mathrm{temp}}$ denotes the threshold for accepting worse solutions and $\mathrm{fit}_{\mathrm{Gbest}}$ denotes the EAoI_q value corresponding to the globally optimal solution.

C. Optimizing $\{p_k\}$ for Given $\{\mathbf{q}_s\}$ and $\Theta_{s,k}$

After obtaining the position of the UAV's stopping point $\{\mathbf{q}_s\}$ and the closed expression of $\Theta_{s,k}$, the transmit powers optimization problem of IoT devices is also independent of the first and third terms of $(\mathcal{P}_0)$, thus it can be expressed by EAoI_p, which can be expressed as

$$(\mathcal{P}_{3.1}):\ \min_{\{p_k\}}\ \mathrm{EAoI}_p = \mathrm{EAoI}_{\Theta_{s,k}} \quad \text{s.t. (20c)} \tag{33}$$

where the result of EAoI_p represents the effect of the optimization of the transmit powers on EAoI.

It can be seen that the problem $(\mathcal{P}_{3.1})$ cannot be solved directly because of its nonconvexity. For ease of handling, thus we let $\tau = (|h_{s,k}^U|^2/[\sum_{j=1}^J p_j|h_{j,s}^{J,U}|^2 + \sigma_s^2]) > 0$, thereby expressing (9) as

$$R_{sk} = \hat{B}\log_2(1 + \tau p_k). \tag{34}$$

Then, we convert the problem $(\mathcal{P}_{3.1})$ into the following formula:

$$(\mathcal{P}_{3.2}):\quad \max_{\{p_k\}}\ -\mathrm{EAoI}_p \quad \text{s.t. (20c)}. \tag{35}$$

At this point, we can approximate using the SCA method [46]. Specifically, the surrogate function, denoted by $\mathcal{F}(p_k|p_k^{\langle c,T\rangle})$, is required to satisfy the following features, including 1) function value consistency; 2) gradient consistency; 3) an upper-bound; and 4) convexity, which can be represented as follows:

$$\mathcal{F}\left(p_k^{\langle c,T\rangle}|p_k^{\langle c,T\rangle}\right) = F\left(p_k^{\langle c,T\rangle}\right) \tag{36a}$$

$$\nabla\mathcal{F}\left(p_k^{\langle c,T\rangle}|p_k^{\langle c,T\rangle}\right) = \nabla F\left(p_k^{\langle c,T\rangle}\right) \tag{36b}$$

$$\mathcal{F}\left(p_k|p_k^{\langle c,T\rangle}\right) \le F(p_k) \tag{36c}$$

where $p_k^{\langle c,T\rangle}$ denotes the obtained optimal solution in the cth inner iteration and the Tth outer iteration, and $\nabla F(p_k^{\langle c,T\rangle})$ denotes the gradient at $p_k^{\langle c,T\rangle}$. The convexity requirement can be satisfied by constructing the surrogate function using a Taylor expansion as follows:

$$F(p_k) \ge F\left(p_k^{\langle c,T\rangle}\right) + \nabla F\left(p_k^{\langle c,T\rangle}\right)\left(p_k - p_k^{\langle c,T\rangle}\right) \tag{37}$$

where we can use the result of the previous iteration p_k' to replace $p_k^{\langle c,T\rangle}$. Then, after Taylor expansion, we can get the following two inequalities:

$$-\frac{1}{R_{jk}} \ge -\frac{1}{\log_2\left(1+\tau p_k'\right)} + \frac{1}{\ln 2}\frac{\tau\left(p_k - p_k'\right)}{\left(1+\tau p_k'\right)\left[\log_2\left(1+\tau p_k'\right)\right]^2} \tag{38}$$

$$-\frac{p_k}{R_{jk}} \ge -\frac{p_k'}{\log_2\left(1+\tau p_k'\right)} + \left\{-\frac{\left(p_k - p_k'\right)}{\log_2\left(1+\tau p_k'\right)} + \frac{\tau p_k'\left(p_k - p_k'\right)}{\ln 2\left(1+\tau p_k'\right)\left[\log_2\left(1+\tau p_k'\right)\right]^2}\right\}. \tag{39}$$

So we can finally represent the problem $(\mathcal{P}_{3.1})$, shown at the bottom of the next page

Therefore, the problem $(\mathcal{P}_{3.3})$ becomes convex and can be solved efficiently with CVX. In the solution process if the transmit powers of all the IoT devices is solved directly then various combinations of the transmit power of these devices need to be considered, in this case when the number of IoT devices increases the combinations of the power will also increase and the computational complexity will grow exponentially. Therefore, to reduce the complexity of the solution, we adopt the method of segmented storage and joint solution, that is, the three items in the problem $(\mathcal{P}_{3.3})$

Algorithm 2: SCA Algorithm for Solving Problem $\mathcal{P}_{3.3}$

Initialization: Power p_{k0} of IoT devices, $c = 1$;
repeat
  Given $p'_k = p_{k0}$, the intermediate solution p_k^c is obtained by solving problem $(\mathcal{P}_{3.3})$ through CVX;
  Update $p'_k = p_k^c$, $c \leftarrow c + 1$;
until The fractional increase of the objective value is less than ε_{SCA} or $c > \hat{c}$;
return p_k.

Algorithm 3: Overall Algorithm for Solving Problem $\mathcal{P}_0$

Initialization: $T = 1$, $\Theta_{s,k}$, $\mathbf{q}_{s0}$, p_{k0}, P_J, BS position $\mathbf{q}_B$, jammer position $\mathbf{q}_J$, IoT devices position $\mathbf{q}_k$;
$\mathbf{q}_s^* = \mathbf{q}_s^T = \mathbf{q}_{s0}$, $p_k^* = p_k^T = p_{k0}$;
Closed expression for solving diagonal phase shift matrix $\Theta_{s,k}$ of IRS.
repeat
  The JPACA algorithm is used to solve alternately, including:
  1) Given p_k^* and closed expression of $\Theta_{s,k}$, $(\mathcal{P}_2)$ is solved by the VPPSA algorithm to obtain solution $\mathbf{q}_s^{T+1}$, set $\mathbf{q}_s^* \Leftarrow \mathbf{q}_s^{T+1}$;
  2) Given $\mathbf{q}_s^*$ and closed expression of $\Theta_{s,k}$, $(\mathcal{P}_{3.3})$ is solved by the SCA algorithm to obtain solution p_k^{T+1}, set $p_k^* \Leftarrow p_k^{T+1}$.
  Calculate EAoI according to the results of the JPACA algorithm, and the diagonal phase shift matrix of IRS, the UAV trajectory and the transmit powers of IoT devices corresponding to min-EAoI were recorded by comparison;
  Update $T \leftarrow T + 1$;
until The fractional increase of EAoI is less than ε_{OA} or $T > T_{max}$;
return EAoI, $\Theta_{s,k}$, p_k^{**}, and $\mathbf{q}_s^{**}$.

are constructed separately, and the optimization results of the transmit powers of IoT devices in each item are stored separately according to the device index. In the final solution process, the items are combined to obtain the suboptimal solution of the problem. This can avoid a large increase in computational complexity when the number of IoT devices is large. Therefore, the proposed SCA-based algorithm is summarized in Algorithm 2, where $p_{k0}\ \ \forall k \in \mathcal{K}$ denotes the initial transmit power of the IoT devices, and $0 < \varepsilon_{\text{SCA}} \ll 1$ denotes the convergence tolerance.

D. Overall Solution

After obtaining a closed expression for the diagonal phase shift matrix of the IRS, the suboptimal solution of the JPACA algorithm is obtained by alternating problem $(\mathcal{P}_2)$ and problem $(\mathcal{P}_3)$ optimizations and the details of the overall solution are summarized in Algorithm 3, where $\mathbf{q}_{s0}$ and p_{k0} are the initial values of the given stopping point position and the transmit power of IoT devices, respectively, $\mathbf{q}_s^*$ and p_k^* are the suboptimal solutions of all iterative processes so far, and $0 < \varepsilon_{AO} \ll 1$ indicates the collection tolerance. Then, we use $\mathbf{q}_s^{**}$ and p_k^{**} to represent the solution of the last iteration. The convergence and complexity analysis are provided as follows.

1) Convergence Analysis: First, in the VPPSA algorithm, because we only consider the introduction of the SA algorithm in the update of individual optimal solutions, we can ignore its effect on the global optimal solution. Moreover, we make the solution after each iteration the initial global optimal solution in the next iteration, that is, the VPPSA algorithm of each iteration is optimized based on the last one, thus we can get

$$\text{EAoI}\left(\Theta_{s,k}, p_k^*, \mathbf{q}_s^T\right) \geq \text{EAoI}\left(\Theta_{s,k}, p_k^*, \mathbf{q}_s^{T+1}\right) \tag{41}$$

where the inequality holds since a new set of positions in our algorithm can replace the original global optimal solution only when the value of the fitness function is less than the value of the global optimal solution, it is guaranteed that VPPSA eventually converges to the optimal solution. The internal iteration process of the algorithm is the same, and the internal iteration will not fall into the local optimal solution with the help of the SA algorithm. Second, when solving problem $(\mathcal{P}_3)$ by SCA method, according to (36), (38), and (39), we have

$$\begin{aligned} F\left(p_k^{\langle c-1,T\rangle}\right) &= \tilde{F}\left(p_k^{\langle c-1,T\rangle} | p_k^{\langle c-1,T\rangle}\right) \\ &\leq \tilde{F}\left(p_k^{\langle c,T\rangle} | p_k^{\langle c-1,T\rangle}\right) \leq F\left(p_k^{\langle c,T\rangle}\right) \end{aligned} \tag{42}$$

where the above inequality holds, because when $\Theta_{s,k}$ and $\mathbf{q}_s^*$ are given, we obtain a suboptimal solution for problem $(\mathcal{P}_3)$. This shows that Algorithm 2 does not decrease after each iteration and eventually converges to the maximum value. According to Cauchy's theorem [47], since $\mathbf{q}_s$ and p_k are both bounded by constraints, the overall algorithm is guaranteed to have a monotonic convergence.

2) Complexity of the algorithm: In the case of obtaining the closed expression of the diagonal phase shift matrix and given p_k, the time complexity of the VPPSA algorithm in the worst case is $\mathcal{O}(2NPS + NP)$. Similarly, in the case of given $\mathbf{q}_s$, since the SCA algorithm updates the transmit power of all K IoT devices in each iteration, the time complexity of the SCA algorithm in the worst case is $\mathcal{O}(\hat{c}K^3)$. So the total computational complexity is $\mathcal{O}(T(2NP(\lceil[K/W]\rceil + [1/2]) + \hat{c}K^3))$.

$$\begin{aligned} (\mathcal{P}_{3.3}) : \max_{\{p_k\}} \ & -\frac{\omega_2 \zeta D}{\hat{B}} \sum_{s=1}^{S} \sum_{j=s}^{S} \sum_{k=w_{j-1}+1}^{w_j} \frac{1}{\log_2\left(1+\tau p'_k\right)} - \frac{1}{\ln 2} \frac{\tau}{\left(1+\tau p'_k\right)\left[\log_2\left(1+\tau p'_k\right)\right]^2} \left(p_k - p'_k\right) \\ & - \frac{(1-\omega_2) p_h D}{\hat{B}} \sum_{s=1}^{S} \sum_{k=w_{s-1}+1}^{w_s} \frac{1}{\log_2\left(1+\tau p'_k\right)} - \frac{1}{\ln 2} \frac{\tau}{\left(1+\tau p'_k\right)\left[\log_2\left(1+\tau p'_k\right)\right]^2} \left(p_k - p'_k\right) \\ & - \frac{\omega_1 (1-\omega_2) D}{\hat{B}} \sum_{s=1}^{S} \sum_{k=w_{s-1}+1}^{w_s} \frac{p'_k}{\log_2\left(1+\tau p'_k\right)} + \left\{ \frac{1}{\log_2\left(1+\tau p'_k\right)} - \frac{\tau p'_k}{\ln 2\left(1+\tau p'_k\right)\left[\log_2\left(1+\tau p'_k\right)\right]^2} \right\} \left(p_k - p'_k\right) \\ \text{s.t.} \ & \text{(20c)} \end{aligned} \tag{40}$$

TABLE I
SIMULATION PARAMETER

Parameters	Descriptions	Values
H_0	Flight height of the UAV	200 m [33]
p_k^{max}	Maximum transmit power of IoT devices	30 dBm
P_J	Transmit power of jammer	20 dBm
p_f	Flight power of UAV	1283 W [29]
p_h	Hover power of UAV	1000 W [29]
M_x, M_y	The number of reflecting elements	5, 10
D	The amount of data	5×10^3 MB
W	Maximum number of served devices	5 [29]
V	The flight speed of the UAV	50 m/s [30]
$\hat{B}$	System bandwidth	1 MHz [31]
α	Constants related to the practical environment	12.08 [31]
β	Constants related to the carrier frequency	0.11 [31]
σ^2	Noise power	-200 dBm
γ_1, γ_2	Path loss exponent	0.01
d	IRS element separation distance	$\lambda/2$
ω_1, ζ	Compensation factor	2×10^3, 500
ω_2	Weight coefficient	0.5
K	Default number of IoT devices	30
J	Default number of Jammers	1
$N, \hat{c}, T_{max}$	Maximum iteration number	200, 20, 100
T_{ini}	The initial temperature	4000
t	Coefficient of curve change rate	1.8

TABLE II
ALGORITHM PARAMETER SETTING

Algorithm	Parameter	Reference
VPPSA	$w_{max} = 2, w_{min} = 0.4, c_1 = 1.6, c_2 = 2$	\
PSO	$w_{max} = 2, w_{min} = 0.4, c_1 = 1.6, c_2 = 2$	[48]
GWO	$\mu = 0.4, a = 2 \sim 0$	[49]
SDPSO	$w_{max} = 0.9, w_{min} = 0.4, c_1 = 1.6, c_2 = 2$	[28]
AWOA	$\mu = 4, a = 2 \sim 0, \rho = 0.01$	[29]
JPACA*	$w_{max} = 2, w_{min} = 0.4, c_1 = 1.6, c_2 = 2$	\
TPSO	$w_{max} = 2, w_{min} = 0.4, c_1 = 1.6, c_2 = 2$	\

IV. PERFORMANCE EVALUATION

A. Simulation Setup

In this section, the network performance is evaluated by simulation results, and the effectiveness of the proposed algorithm is proved. All the IoT devices were placed in a square area of 1000m $\times$ 1000m according to Poisson distribution, and only one UAV was used to collect all the data of IoT devices, i.e., $x_s^{\max} = y_s^{\max} = 1000 \quad \forall s \in \mathcal{S}$. At the same time, as shown in Fig. 1, we assume that the locations of BS, IRS, and Jammer are (1000, 500, 100), (500, 0, 100), and (500, 500, 100), respectively. The other parameters in the system model are shown in Table I, where $T_{\max}$ is the maximum number of iterations of the JPACA algorithm and each iteration includes a complete loop of trajectory optimization and power optimization. The N and $\hat{c}$ are the maximum iterations of the VPPSA and the SCA algorithm, respectively.

The relevant parameters of the algorithms involved in the comparison are shown in Table II, in which the VPPSA algorithm is the algorithm proposed in this article, the PSO [48] algorithm is the baseline algorithm, the grey wolf optimizer (GWO) [49] algorithm is the baseline trajectory optimization algorithm, novel hybrid PSO (SDPSO) [28] is the PSO algorithm that introduces SA algorithm in the process of updating the global optimal solution, adaptive whale optimization algorithm (AWOA) [29] is the optimization algorithm to minimize the AoI, and the JPACA* algorithm is a version that changes the objective function of the algorithm proposed in this article from (20a) to (19). TPSO algorithm is a PSO algorithm that introduces random disturbance to the particle's position x_i^n after each iteration to avoid falling into the local optimal solution, where the disturbance threshold is dt, the disturbance amplitude is dr, and the disturbance process is shown as follows:

$$x_i^n = \begin{cases} x_i^n + dr, & \text{if} \quad \text{rand} < dt \\ x_i^n, & \text{others} \end{cases} \tag{43}$$

where $dt = 0.06$, $dr = 0.1$, and *rand* represents the generated random number between 0 and 1. Meanwhile, the transmission rate in the cases where there are jammers in the system but there is no anti-jamming optimization as follows:

$$R'_{sk} = \hat{B} \log_2\left(1 + \frac{p_k |h_{s,k}^U|^2}{\sigma_s^2}\right). \tag{44}$$

In this article, the closed-form solution of subproblem $(\mathcal{P}_{1.2})$ is obtained, and the formula of the suboptimal solution of subproblem $(\mathcal{P}_{3.3})$ is also obtained. Therefore, we mainly demonstrate the convergence of VPPSA, JPACA, and SCA algorithms, and simulate the performance of VPPSA and JPACA algorithms.

B. Convergence Performance

The convergence analysis of the VPPSA algorithm and the JPACA algorithm proposed in this article is shown in Fig. 4. It can be seen that the convergence of the VPPSA algorithm is the same as that of most heuristics, and the number of iterations will increase with the increase of particle complexity, but eventually, they can all converge around a value. Therefore, the number of iterations can be reduced by increasing the population number according to demand. In addition, it can be seen from Fig. 4(c) and (d) that the convergence of the JPACA algorithm based on the AO algorithm is faster because the convergence of the VPPSA algorithm and the SCA algorithm is good under the parameters we set. This also shows that the algorithms proposed in this article have good convergence.

It can be seen that the solution to the problem $(\mathcal{P}_{3.3})$ is not only limited by (20c) but also by the number of IoT devices. Fig. 5 is a figure obtained by convex approximation using segmented storage and joint solution when the number of IoT devices is 50, where the asterisk indicates the optimal transmit power of the IoT device. The reason for the smaller $-\text{EAoI}_p$ value at two points in the figure is that these two devices are located in relatively remote locations, thus the importance of these two devices is weakened to ensure the minimization of the overall EAoI during the trajectory planning process. The figure clearly shows the relationship between the target function value and the number of IoT devices and the transmit power and also proves the rationality and optimality of the transmit power obtained under the constraints of the original function nonconvex and in (20c).

C. Demonstration of VPPSA Algorithm

Fig. 6 is a distribution scatter plot of IoT devices, BS, IRS, jammers, and optimized stopping points. The UAV starts from the BS, passes through all the stops in turn, returns to the

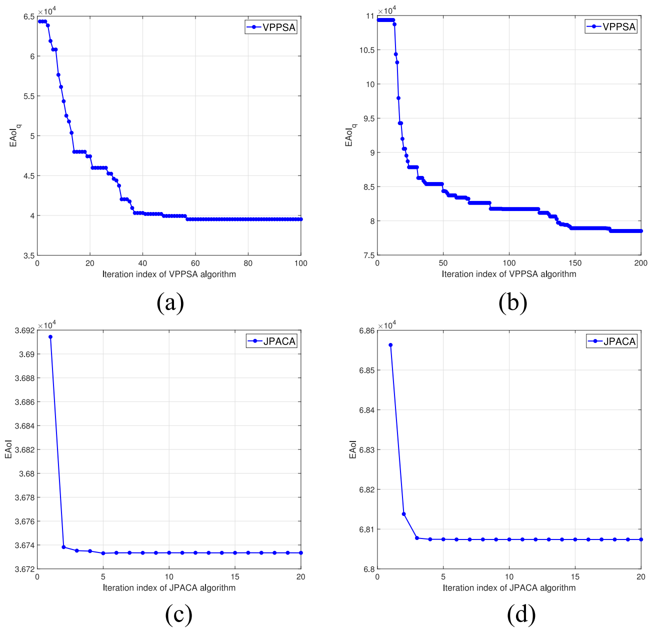

Fig. 4. Convergence performance of the VPPSA algorithm and JPACA algorithm. (a) $K = 30$. (b) $K = 50$. (c) $K = 30$. (d) $K = 50$.

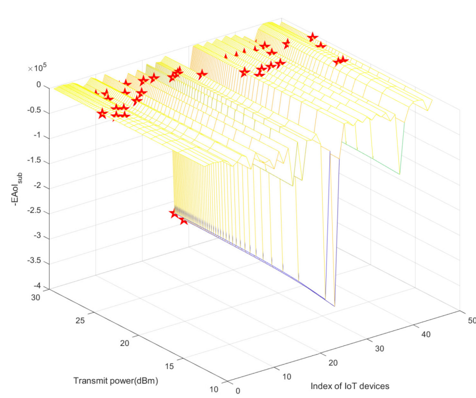

Fig. 5. Correlation between objective function value and IoT device quantity and power.

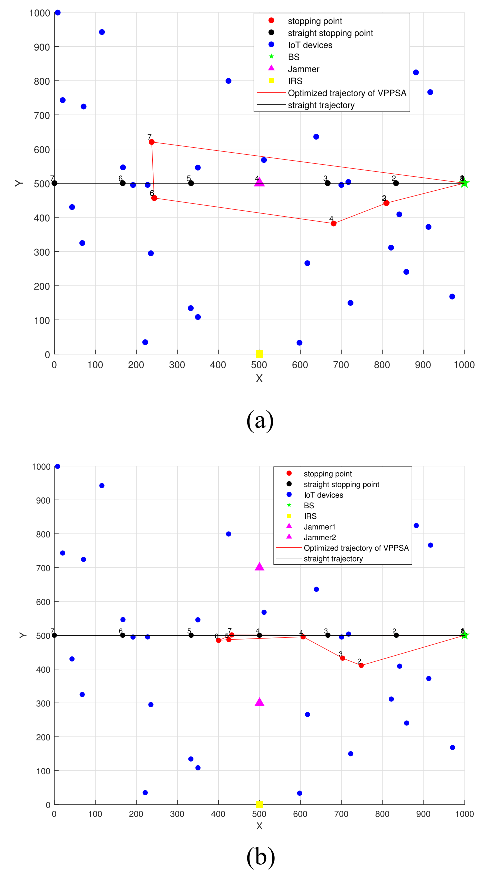

Fig. 6. Trajectory comparison and demonstration. (a) UAV trajectory diagram optimized by VPPSA algorithm. (b) Optimized trajectory of UAV diagram with two jammers.

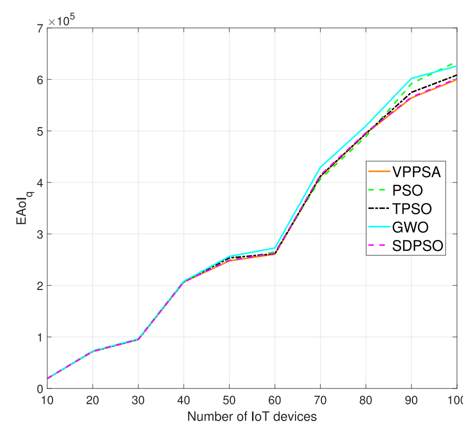

Fig. 7. EAoI performance for different numbers of IoT devices.

BS, and sends the data back to the BS. Therefore, the VPPSA algorithm controls the distance between the UAV and the IoT devices and between the UAV and the jammers by adjusting the position of the stopping point and adjusting the flight distance to achieve anti-jamming and minimize EAoI. From Fig. 6(a) we can see that different trajectories lead to different flight distances and times, which also affects the transmit power of the IoT device. In addition, we can also see that the distance between the second and third stopping points of the UAV is relatively close since there is a relative concentration of IoT devices that need to communicate at these two stopping points. From Fig. 6(b), it can be seen that when only the number and position of jammers in the network change, the UAV trajectory will also change accordingly, which shows that the algorithm proposed in this article allows the UAV to adjust its flight trajectory according to the actual situation when the network state caused by jamming changes, thereby reducing the effect of jamming on communication.

Fig. 7 shows the performance comparison of the VPPSA algorithm, PSO algorithm, and several other optimization algorithms under different numbers of IoT devices, where the results under each number are the average of 100 different distribution results. As can be seen from the figure, although the performance of the VPPSA algorithm is the best when the number of IoT devices is small, the performance of various algorithms is not different. This is because the overall complexity is relatively small, and the algorithm cannot easily fall into the local optimal solution. With the increase in the number

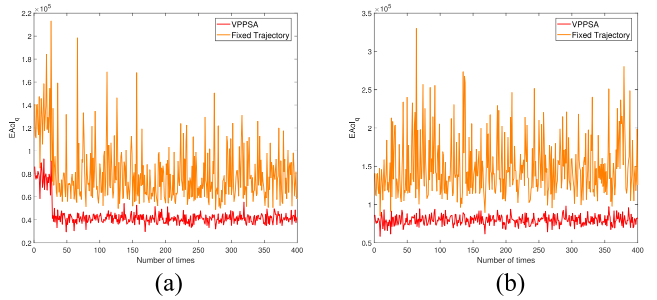

Fig. 8. Trajectory optimization performance demonstration. (a) Chang of EAoI when $K = 30$. (b) Chang of EAoI when $K = 50$.

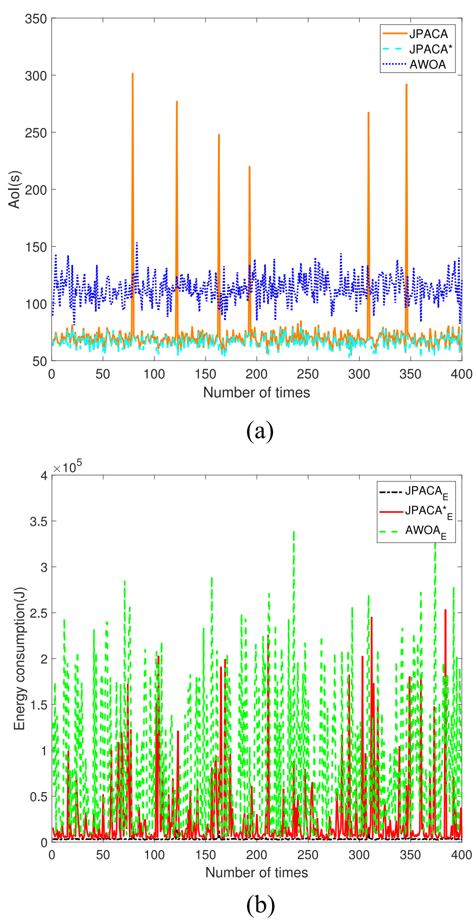

Fig. 9. Optimized performance demonstration of AoI and EC. (a) AoI performance comparison. (b) EC performance comparison.

of devices, the advantages of the VPPSA algorithm, TPSO algorithm, and SDPSO algorithm gradually become obvious and the VPPSA algorithm has the best effect. At the same time, the SDPSO algorithm also needs to use a dimensional learning strategy (DLS) to reduce particle fluctuations. This shows that the algorithm proposed in this article can adapt well to the environment of different devices, even if the number of devices in the environment suddenly increases, the algorithm can timely adjust the UAV trajectory to cope with such changes.

Fig. 8 shows the effect of the distribution of devices and the trajectory of UAV on EAoI under different IoT device

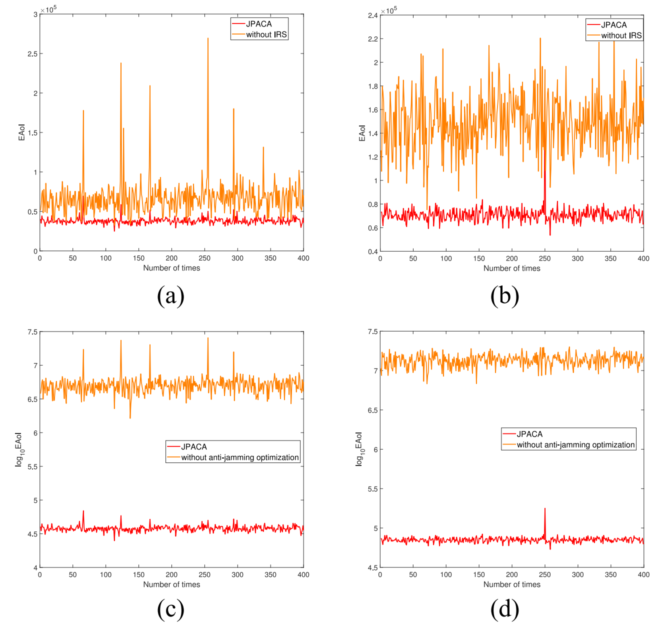

Fig. 10. Optimized IRS and anti-jamming performance demonstration. (a) $K = 30$. (b) $K = 50$. (c) $K = 30$. (d) $K = 50$.

numbers. It can be seen from the figure that the impact of the trajectory of UAV on EAoI shows the same trend when the number of devices is different, and the change of device distribution has a greater impact on the EAoI corresponding to the fixed trajectory. The mutual confirmation with Fig. 7 shows that the algorithm proposed in this article has better average performance in trajectory optimization.

D. Demonstration of Overall Algorithm

Fig. 9 shows the performance comparison results of AoI and EC under different IoT device distributions. It can be seen from Fig. 9(a) that the AoI performance index of the JPACA* algorithm is the best under each distribution. The AoI metric of the JPACA algorithm is close to the JPACA* algorithm and is better than the AOWA algorithm. Several outliers appear in the figure because the distribution is similar and the devices are close to the jammer. To better balance the relationship between AoI and EC, AoI will inevitably increase. It can also be seen from Fig. 9(b) that the EC of the JPACA algorithm optimization is smooth and minimal, and the EC of the JPACA* algorithm is close to that of the AOWA algorithm but better than that of the AOWA algorithm. This shows that the JPACA algorithm outperforms the AOWA algorithm in minimizing AoI on average, can balance the relationship between AoI and EC, and effectively reduces both metrics to improve system performance.

Fig. 10 shows the effect of IRS optimization and anti-jamming performance under different device distributions. As can be seen from Fig. 10(a) and (b), the optimized IRS significantly improves both effect and stability, which shows that our optimization method for IRS makes it have better average performance, and can maximize the use of IRS to improve communication rate, thereby reducing AoI. From Fig. 10(c) and (d), it can be seen that whether the

TABLE III
AVERAGE PERFORMANCE COMPARISON OF IRS AND ANTI-JAMMING

Algorithm	Number of device	Mean Value
JPACA	30	3.79×10^4
without IRS	30	6.73×10^4
without anti-jamming optimization	30	5.22×10^6
JPACA	50	7.09×10^4
without IRS	50	1.36×10^5
without anti-jamming optimization	50	1.49×10^7

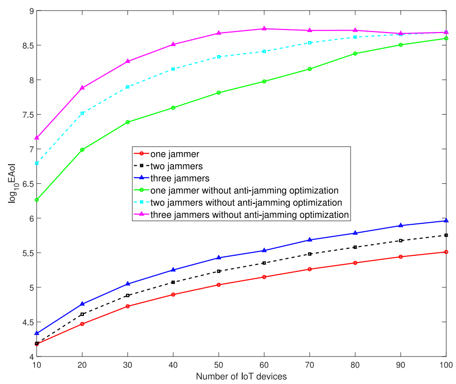

Fig. 11. Anti-jamming performance with different number of IoT devices.

presence of malicious jammers is considered in the trajectory optimization has a great effect on performance. There are two main reasons. First, because the stopping point of the UAV is very close to the actual jammer. The second is that the communication rate calculation error makes the device use inappropriate transmit power, and the superposition of the two eventually leads to a significant increase in EAoI. The average of EAoI in Table III also confirms the point made in Fig. 10. Therefore, the algorithm proposed in this article has good anti-jamming performance and can adapt to more complex and harsh communication environments.

Fig. 11 shows the effect of the numbers of jammers J and IoT devices K on EAoI in (20a). For any J and K, the proposed scheme has lower EAoI than the benchmark scheme, because it can decrease $h_{j,s}^{J,U}$ by adjusting the UAV trajectory and increase $h_{s,k}^{U}$ by adjusting the transmit powers of IoT devices, while the benchmark scheme ignores the effect of jammers. With the increase of K, the EAoI of the proposed scheme increases, since the total load of data collection increases, and this consumes more time and energy. In addition, its EAoI also increases with the increase of J, because the increase of the number of jammers in (9) leads to the decrease of R_{sk}, and thus increases $(T_{\text{fly}} + T_{\text{hov}})$ and E.

V. CONCLUSION

This article studied the problem of anti-jamming and minimization of AoI and EC during data collection in the IRS-assisted UAV-IoT networks in the presence of malicious jamming. To solve the coupled nonconvex problem, it is decomposed into three subproblems for solving based on the AO algorithm. First, the IRS diagonal phase shift matrix optimization problem is solved by the quantitative passive beamforming method, and a closed expression for the phase shift matrix is obtained. Then, the JPACA algorithm is proposed to optimize the UAV trajectory and the transmit powers of the IoT devices. In the JPACA algorithm, a VPPSA algorithm is proposed to solve the trajectory optimization problem of the UAV, and the SCA method is used to solve the transmit powers optimization problem of IoT devices. Finally, the convergence of the proposed algorithm is verified. Meanwhile, simulation results show that the average performance of the scheme in AoI, EC, and anti-jamming of various devices is better than that of the benchmark scheme. In the future work, we aim to consider the data outdating and data collection frequency problems in the EAoI optimization under jamming attacks.

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Peng Wang (Graduate Student Member, IEEE) received the B.S. degree from the School of Computer Science and Technology, Shandong University of Technology, Shandong, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Electronic Information Engineering, Beihang University, Beijing, China.

His research interests include wireless communication, anti-jamming techniques, and UAV communications and networking.

Kai Liu (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Xidian University, Xi'an, China, in 1994, 1997, and 2001, respectively.

From March 2000 to February 2001, he was a Visiting Researcher with Shizuoka University, Hamamatsu, Japan. From January 2002 to February 2004, he was a Senior Research Associate with Illinois Institute of Technology, Chicago, IL, USA. From February 2015 to March 2016, he was a Visiting Scholar with Texas A&M University, College Station, TX, USA. He is currently a Professor with the School of Electronics and Information Engineering, Beihang University, Beijing, China. His research interests include information communication technology and computing intelligence.

Prof. Liu is a Senior Member of CIE.

Yaodong Ma (Graduate Student Member, IEEE) received the B.S. degree in communication engineering from Ningbo University, Ningbo, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Electronics and Information Engineering, Beihang University, Beijing, China.

His research interests include wireless communication, game theory, physical-layer security, and UAV communications and networking.

Qiang Gao (Senior Member, IEEE) received the B.Sc. degree in physics from Southwest China Normal University, Chongqing, China, in 1994, the M.Sc. degree in physics from Lanzhou University, Lanzhou, China, in 1997, and the Ph.D. degree in computer from Chinese Academy of Sciences, Beijing, China, in 2000.

From 2001 to 2004, he worked as a Postdoctoral Researcher with the National Research Center of Information Technology, Bonn, Germany. He was a Research Fellow with Aston University, Birmingham, U.K. He joined Beihang University, Beijing, as an Associate Professor in 2004, where he is currently a Professor with the School of Electronic and Information Engineering. He is the author, co-author, and an editor of three books and more than 80 technical articles. His research interests include machine learning, wireless communication, and wireless networks.

Dr. Gao is a winner of the Chinese Education Ministry's New Century Excellent Talents Supporting Plan. He is a Senior Expert of the Regional Center for Space Science and Technology Education in Asia and the Pacific. He is a Senior Member of the China Computer Federation.